

Kelly Cao

keca4089@gmail.com | 720-382-9093 | LinkedIn: <https://www.linkedin.com/in/kelly-cao-490a031b8/>

Education

Purdue University – West Lafayette, IN

January 2025 – May 2026

Master of Science in Biomedical Engineering, Biomedical Device Development Concentration

Graduate Certificate: Regulatory Affairs and Regulatory Science For Medical Devices

University of Colorado Boulder – Boulder, CO

August 2020 – May 2024

Bachelor of Science in Biomedical Engineering

Minor: Computational Biology

Technical Skills

- **Programming:** Python, MATLAB, SQL, C++
- **Design:** SolidWorks, FEA, CFD simulation, 3D modeling and visualization, GD&T, reverse engineering
- **Software:** Google Docs/Sheets, Microsoft Word/Excel/Teams, Slack, Adobe InDesign

Relevant Experience

Reef-Safe Personal Care Product Internship – Magwai

May 2025 – Sep 2025

- Supported early product design and development by conducting comprehensive market reviews to understand current product trends and identify key research areas for marine safe products.
- Collaborated with the product team to propose innovative reef-safe formulations and ingredient solutions for early product development.
- Gathered inputs on potential toll manufacturers and packaging suppliers to help the product team make the most sustainable and beneficial decisions for new products.

Entry BioCAD Tech - 3D Systems

May 2023 – August 2023

- Utilized SolidWorks to design patient-specific orthognathic splints for oral maxillofacial surgical procedures.
- Developed detailed case reports consisting of images of orthognathic splints with relevant anatomy and surgeon preferred images.
- Achieved daily objectives in a timely manner within a fast-paced work environment.

Computational Flow Dynamics (CFD) Research Assistant - University of Colorado Boulder

June 2021 – May 2024

- Designed patient-specific CFD models for LVAD patients, Glenn/Fontan procedure patients, and embolic stroke patients.
- Leveraged CFD simulation results to visualize and understand hemodynamics in the modeled regions of interest (ROIs).
- Analyzed medical imaging data like 3DRA or CT imaging for proper segmentation of ROIs during the model creation process.
- Communicated with team members in producing patient-specific cardiovascular simulations.
- Produced and managed simulation data and created technical presentations and documentations.

Research Experience

Graduate Data Science Researcher – The Data Mine

August 2025 – December 2025

- Collaborated with Johnson & Johnson's Innovative Medicine Hematology team to create a project dashboard identifying trends and metrics of the performance of continuing medical education programs.
- Programmed an automated dashboard for stakeholders using Python and PowerBI skills.

Projects

Quality Control Project

January 2025 – May 2025

- Applied the DMAIC framework and utilized statistical quality control tools like Pareto charts, np charts, and time series plots to identify causes of item defects.
- Performed root cause analysis with a cause-and-effect diagram to suggest potential process modifications for continuous improvement.

Dispensing and Dosing System Project

August 2023 – May 2024

- Contributed to sustaining engineering efforts by optimizing previous design iterations of a dispensing and dosing system for bone graft.
- Modified component dimensions and added new components to help automate the system.
- Conducted iterative testing to improve dosing efficiency in volume and time, meeting dosing size requirements and increasing production rate.
- Created toleranced computer-aided design (CAD) SolidWorks drawings and prepared technical reports and presentations for clients and advisors.

Relevant Coursework

- **Wearable Sensors:** analyzed personal data collected from a monitoring bracelet to identify different trends in heart, sleep, and breathing.
- **Regulatory Approval:** created an FDA pre-submission packet with a team and learned about various submission pathways.
- **Quality Systems:** obtained knowledge about topics related to FDA's Quality System Regulation and ISO 13485.
- **Lean Manufacturing:** researched and synthesized information about the different types of waste and manufacturing environments for lean manufacturing and pharmaceutical product development.

Certifications

- **Six Sigma Green Belt**, Institute of Industrial and Systems Engineers (IISE), April 2026
- **Engineering Leadership Fundamentals**, Purdue University, 2026